

INTERMEDIATE EXAMINATION -2019

(ANNUAL)

Time- 31/4 Hours

Full Marks: 70

Instruction for the candidates:

- (1) Candidates are required to give their answer in their own words as far as practicable.
- (2) Figure in the right hand margin full marks.
- (3) 15 Minutes of extra time has been allowed for the candidates to read the questions carefully.
- (4) This question paper is divided into two sections- **Section-A and Section-B**
- (5) In **Section-A**, there are 28 **objectives types question** which are compulsory. Each carrying **1 mark**. Darken the circle with blue black pen against the correct option on OMR Sheet provided to you.
Do not use whitener/Liquid Blade/Nail on OMR Paper, otherwise the result will be invalid.
- (6) In **Section-B**, there are **II short answer type question (each carrying 2 marks)**, out of which **any 11 questions are to be answered. Apart from this, there are 4 long answer Type questions. (Each Carrying 5 marks)**, out of which **any 3 question are to be answered.**
- (7) Use of any electronic device is prohibited.

1. The electric dipole moment of an electric dipole made up of two opposite charges have magnitude $+3.2 \times 10^{-19} C$ and $-3.2 \times 10^{-19} C$ separated by a distance $2.4 \times 10^{-10} m$ is
 (A) $7.68 \times 10^{-27} C - m$ (B) $7.68 \times 10^{-29} C - m$
 (C) $7.86 \times 10^{-29} C - m$ (D) $7.86 \times 10^{-27} C - m$

Answer: (B)

2. Identical drops each of capacity $5 \mu F$ combine to form a big drop. what is the capacity of big drop?
 (A) $16 \mu F$ (B) $20 \mu F$ (C) $4 \mu F$ (D) $25 \mu F$

Answer: (B)

3. Kilowatt-hour (k Wh) is the unit of

(A) energy (B) power (C) torque (D) force

Answer: (A)

4. The algebraic sum of all current meeting at any any point in an electrical circuit is

- (A) Infinite (B) positive
(C) Zero (D) negative

Answer: (C)

5. Dimension of permeability is

- (A) $MLT^{-2}I^{-2}$ (B) MLT^2I^{-2}
(C) MLT^2I^2 (D) $MLT^{-2}I$

Answer: (A)

6. Impedance of L-R circuit is

- (A) $R^2 + w^2L^2$ (B) $\sqrt{R + wL}$ (C) $R + wL$ (D) $\sqrt{R^2 + w^2L^2}$

Answer: (D)

7. The direction of propagation of electromagnetic wave is

- (A) parallel to $\vec{B}$ (B) parallel to $\vec{E}$
(C) parallel to $\vec{B} \times \vec{E}$ (D) parallel to $\vec{E} \times \vec{B}$

Answer: (D)

8. When a ray of light enters a glass slab, its wavelength

- (A) increase (B) decrease
(C) remains unchanged (C) data are not complete

Answer: (8)

9. The value of amplitude modulation index is

- (A) Always 0 (B) between 1 and ∞
(C) Between 0 and 1 (D) always ∞

Answer: (C)

10. Boolean expression for NOR gate is

(A) $A + B = Y$

(B) $\overline{A.B} = Y$

(C) $A.B = Y$

(D) $\overline{A + B} = Y$

Answer: (D)

11. The intensity of electric field at any point on the surface of charge conductor is

(A) zero

(B) perpendicular of surface

(C) tangential to the surface

(D) at 45° to the surface

Answer: (B)

12. Permittivity of free space is

(A) $9 \times 10^9 \text{ mF}^{-1}$

(B) $1.6 \times 10^{-19} \text{ C}$

(C) $8.85 \times 10^{-12} \text{ Fm}^{-1}$

(D) $8.85 \times 10^9 \text{ Fm}^{-1}$

Answer: (C)

13. Power of electric circuit is

(A) $V.R$

(B) $V^2.R$

(C) V^2 / R

(D) $V^2.R.I$

Answer: (C)

14. Electron-volt (eV) is the measure of

(A) Charge

(B) potential difference

(C) Current

(D) energy

Answer: (D)

15. Len'z law is associated with

(A) charge

(B) mass

(C) energy

Answer: (C)

16. The final image in astronomical telescope is

- (A) Real and erect (B) real and inverted
(C) Virtual and inverted (D) virtual and erect

Answer: (D)

17. When two converging lenses of same focal length f and concave then the focal length of the combination is

- (A) F (B) $2f$ (C) $\frac{f}{2}$ (D) $3f$

Answer: (C)

18. The transverse nature of light wave supports

- (A) Interference (B) reflection (C) dispersion (D) diffraction

Answer: (C)

19. The colour of a thin film is due to

- (A) Scattering (B) interference (C) dispersion (D) diffraction

Answer: (B)

20. If ϕ is the phase difference between ac current and e.m.f. then the value power of factor is.

- (A) $\tan \phi$ (B) $\cos^2 \phi$ (C) $\sin \phi$ (D) $\cos \phi$

Answer: (D)

21 Unit of self-induction is

- (A) Weber (B) ohm
(C) henry (D) gauss.

Answer: (C)

Answer: (C)

Section-II : (Non-Objective Type Question Question)

Question Nos. 1 to 11 are of short answer type each question carries 2 marks.

[11 × 2 = 22]

1. Define electric dipole moment and write its SI unit.

Answer: Electric dipole moment: It is defined as the product of either charge and the length of electric dipole. It is denoted by $\vec{P}$

$$P = q \cdot 2a$$

S.I Unit: Coulomb metre

2. Write two main features of Laser rays.

Answer:

3. What is the function of second plate in a parallel plate capacitor?

Answer: The charge is induced on the second plate. So potential of first plate decreases and its capacity increases.

4. Write two uses of shunt.

Answer:

- (i) In the protection of coil of galvanometer.
- (ii) Conversion of galvanometer into Ammeter.

5. The charge on a metallic sphere of radius 9cm is $4 \times 10^{-6} \text{C}$. Calculate the potential energy of charge on the conductor.

Answer: Given, $r = 9\text{cm}$

$$P.E. = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} \frac{Q^2}{4\pi\epsilon_0 r}$$

$$Q = 4 \times 10^{-6} C \quad = \frac{1}{2} \times \frac{(4 \times 10^{-6})^2 \times 9 \times 10^9}{9 \times 10^{-2}}$$

$$P.E. = 8 \times 10^{-1} \text{ joule}$$

6. When two resistors are connected in series and parallel, then their equivalent resistors are 16Ω and 3Ω respectively. Find out the resistance of each resistor.

Answer: Let R_1 and R_2 are two resistance

In series $R_1 + R_2 = 16 \dots \dots \dots (i)$

In parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{3} = \frac{R_1 + R_2}{R_1 R_2} = \frac{16}{R_1 R_2}$$

$$R_1 R_2 = 48$$

Now,

$$R_1 - R_2 = \sqrt{(R_1 + R_2)^2 - 4R_1 R_2} = \sqrt{(16)^2 - 4 \times 48}$$

$$R_1 - R_2 = 8 \dots \dots \dots (ii)$$

Solving equation (i) and (ii)

$$R_1 = 12\Omega$$

$$R_2 = 4\Omega$$

7. Name of energy losses in a transformer.

Answer: Energy losses in transfer: (i) Flux losses (ii) Copper losses (iii) iron (iv) Hysteresis loss (V) Humming loss

8. The equation of an ac is $I = 20 \sin 200\pi t$. Find out the frequency. Peak value and r.m.s value of the current.

Answer: Given $I = 20 \sin 200\pi t \dots \dots \dots (i)$

We know that $I = I_0 \sin \omega t \dots \dots \dots (ii)$

Comparing equation (i) and (ii)

$$w = 200\pi$$

$$2\pi f = 200\pi$$

$$f = 100\text{Hz}$$

$$\text{Peak value} = 20\text{A}; I_{\text{rms}} = 0.707I_0 = 0.707 \times 20 = 14.14\text{A}$$

9. Two thin lenses of powers $5D$ and $2D$ are placed coaxially at a distance 20 cm apart. Find the focal length and power of the combination of lenses.

Answer: Power of equivalent lens,

$$P = P_1 + P_2 - dP_1P_2$$

$$5 + 2 - 0.2 \times 5 \times 2$$

$$P = 5D$$

Focal length of equivalent lens

$$f = \frac{1}{p} = \frac{1}{5} \times 100 = 20\text{cm}$$

$$f = 20\text{cm}$$

10. Distinguish between N-type and P-type semiconductors.

Answer: A n-type semiconductor contains a large number of free electron and small number of hole. On the other hand, a P-type semiconductor has a large number of holes and a small number of free electrons.

11. Define modulation. Write its types

Answer: The process of producing a high frequency wave some characteristic of which varies as function of the instantaneous value of audio signal, is called Modulation.

Type of modulation: (i) Amplitude modulation (ii) Frequency modulation (iii) Phase modulation

Question Nos. 12 to 15 are of long answer type. Answer must be explanatory and in your own language. All questions have alternative out of which you have to choose any one alternative.

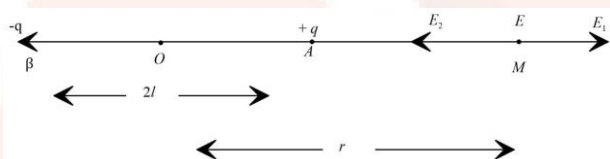
Each question carries 5 marks.

[4 × 5 = 20]

12. Define electric intensity. Derive an expression for electric intensity at a point situated on the axis of electric dipole.

Answer: Electric intensity: The electric field intensity at a point due to source charge may be defined as the force experienced per positive test charge placed at that point without disturbing the source charge.

Expression for electric intensity at a point on the axis of dipole: Let AB is a electric dipole. M is the point at the distance r from its middle point O and A this point expression for electric intensity is to be determined.



The electric intensity at point M due to $+q$ charge along AM

$$E_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(r-l)^2}$$

The electric intensity at point M due to $-q$ charge along MB

$$E_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(r+l)^2}$$

Resultant intensity at point M

$$\begin{aligned} E &= E_1 - E_2 \\ E &= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(r-l)^2} - \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(r+l)^2} \\ E &= \frac{1}{4\pi\epsilon_0} \cdot \frac{q \cdot 4rl}{(r^2 - l^2)^2} \\ E &= \frac{1}{4\pi\epsilon_0} \cdot \frac{2pq}{(r^2 - l^2)^2} \end{aligned}$$

So, l^2 can be neglected.

$$E_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{2p}{r^3}$$

OR

Define electric flux. State and prove Gauss theorem.

Answer:

The number of lines corresponding to a electric field intensity E and surface element ds .

$$d\phi = \vec{E} \cdot d\vec{s}$$

Therefore,

The total flux associated with the whole surface

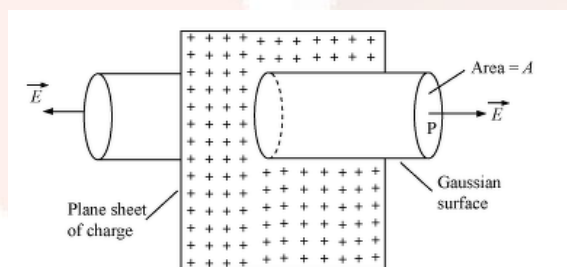
$$\phi = \oint \vec{E} \cdot d\vec{s}$$

Now,

According to Gauss theorem total flux will be equal

$$\phi = \frac{q}{\epsilon_0}$$

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$



Let us consider thin plane sheet of positive charge with a uniform surface charge density σ on both sides of the sheet. Let P be the point at a distance a from the sheet at which the electric field is required.

The electric flux crossing through the Gaussian cylinder,

$$\phi = E \times \text{Area of the circular caps of the cylinder}$$

Since, electric lines of force are parallel to the curved surface of the cylinder, the flux due to the electric field of the plane sheet of charge passes only through the two circular caps of the cylinder.

$$\phi = E \times 2A \dots (1)$$

A/C to Gauss' law

$$\phi = \frac{q}{\epsilon_0}$$

$$q = \sigma A$$

from (1)

$$\therefore \phi = \frac{\sigma A}{\epsilon_0} \dots (2)$$

From eq 1 and eq 2

$$E \times 2 \cdot A = \frac{\sigma A}{\epsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

13. Establish the formula $\frac{\mu_2}{v} - \frac{\mu_2 - \mu_1}{r}$

Answer: Let a concave spherical surface which separates the two medium of refractive index μ_1 & μ_2 .

Let O be a point object in rarer medium and its virtual image is I.

In $\triangle AOC$

$$\alpha = i + \gamma$$

$$i = \alpha - \gamma$$

Since, angle α and γ are small. So,

$$i = \tan \alpha - \tan \gamma = \frac{AN}{NO} - \frac{AN}{NC}$$

As aperture of the spherical surface is small.

$$NO \approx PO; \quad NC \approx PC$$

$$i = \frac{AN}{PO} - \frac{AN}{PC} \dots\dots\dots(i)$$

For $\triangle AIC$ $\beta = r + \gamma; \quad r = \beta - \gamma$

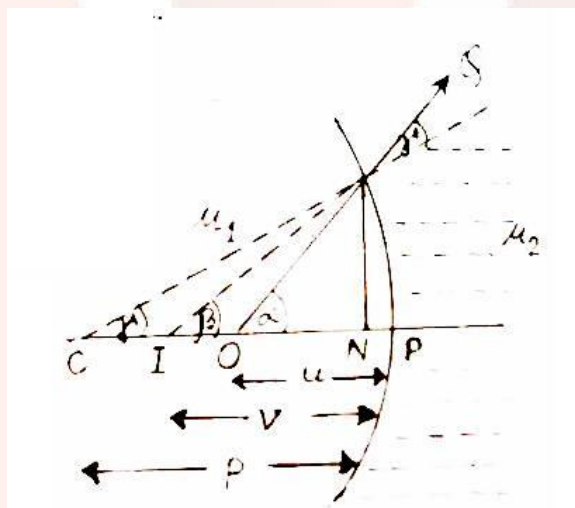
$$r = \tan \beta - \tan \gamma = \frac{AN}{NI} - \frac{AN}{NC}$$

As aperture of spherical surface is small. So,

$$NI = PI$$

$$NC = PC$$

$$r = \frac{AN}{PI} - \frac{AN}{PC} \dots\dots\dots(ii)$$



By Shell's law $\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$
 $\mu_1 \sin i = \mu_2 \sin r$

Since, angle i and r are very small. So, $\mu_1 i = \mu_2 r$

Substituting the value of i and r in above equation

$$\mu_1 \left(\frac{AN}{PO} - \frac{AN}{PC} \right) = \mu_2 \left(\frac{AN}{PI} - \frac{AN}{PC} \right)$$

$$\mu_1 \left(\frac{AN}{PO} - \frac{AN}{PC} \right) = \mu_2 \left(\frac{AN}{PI} - \frac{AN}{PC} \right)$$

$$\mu_1 \left(\frac{1}{PO} - \frac{1}{PC} \right) = \mu_2 \left(\frac{1}{PI} - \frac{1}{PC} \right)$$

$$\mu_1 \left(\frac{1}{PO} - \frac{1}{PC} \right) = \mu_2 \left(\frac{1}{PI} - \frac{1}{PC} \right)$$

$$\mu_1 \left(\frac{1}{-u} - \frac{1}{R} \right) = \mu_2 \left(\frac{1}{-v} - \frac{1}{R} \right)$$

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

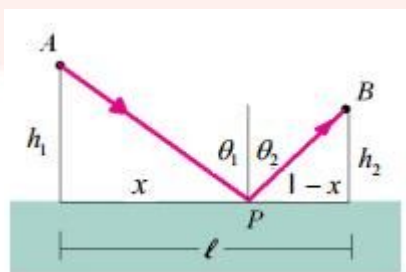
Or,

Write Huygens' Principle of light wave. Prove the law of reflection or refraction of light on the basis of Huygens principle of light of wave.

Answer:

Consider the light ray shown in the figure. A ray of light starting at point A reflects off the surface at point P before arriving at point B, a horizontal distance from point A. we calculate the length of each path and divide the length by the speed of light to determine the time required for the light to travel between the two points.

$$t = \frac{\sqrt{x^2 + h_1^2}}{c} + \frac{\sqrt{(1-x)^2 + h_2^2}}{c}$$



To minimize the time we set the derivative of the time with respect to x equal to zero. We also use the definition of the sine as opposite side over hypotenous to relate the lengths to the angles of incident and reflection.

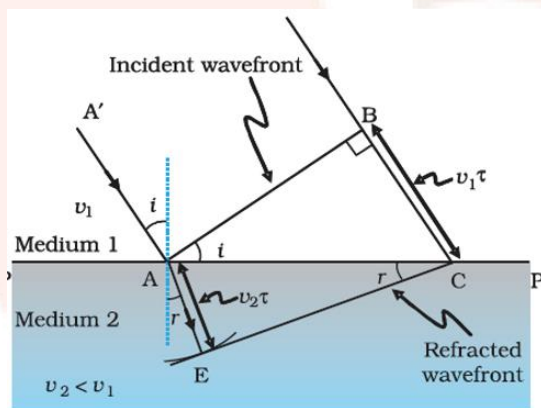
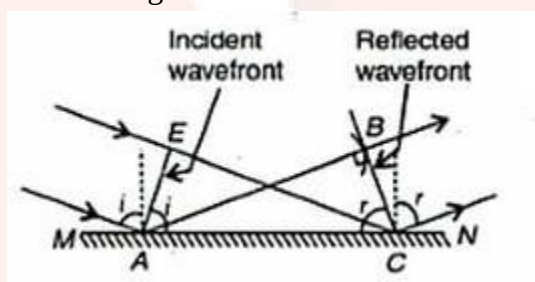
$$0 = \frac{dt}{dx} = \frac{x}{\sqrt{x^2 + h_1^2 c}} + \frac{(1-x)}{\sqrt{(1-x)^2 + h_2^2 c}}$$

$$\frac{x}{\sqrt{x^2 + h_1^2 c}} = \frac{(1-x)}{\sqrt{(1-x)^2 + h_2^2 c}} \rightarrow \sin \theta_1 = \sin \theta_2 \rightarrow \theta_1 = \theta_2$$

Consider a plane wave AB incident at an angle i on a reflecting surface MN. If v is the speed of the wave in the medium and t is the time taken by the wavefront to cover the distance BC, then $BC = vt$

To construct the reflected wavefront we draw an arc (represent reflected wavefront) of radius vt from the point A.

Draw a tangent on the arc. We obtain $AE = BC = vt$



We can see a ray of light is incident on this surface and another ray which is parallel to this ray is also incident on this surface. As these rays are incident from the surface, so we call it incident ray.

Let PP' represent the medium 1 and medium 2. The speed of the light in this medium is represented by v_1 and v_2 . If we draw a perpendicular from point 'A' to this ray of light, Point A, and point B will have a line joining them and this is called as wavefront and this wavefront is incident on the surface.

If ' r ' represents the time taken by the wavefront from the point B to C then the distance,
 $BC = v_1 r$

So, to determine the shape of the refracted wavefront, we draw a sphere of radius v_2r from the point A in the second medium. Let CE represent a tangent plane drawn from the point C on to the sphere. Then, $AE = v_2r$, and CE would represent the refracted wavefront. If we now consider the triangles ABC and AEC, we get that,

$$\sin i = \frac{BC}{AC} = \frac{v_1r}{AC}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2r}{AC}$$

Where, i and r are the angles of incidence and refraction

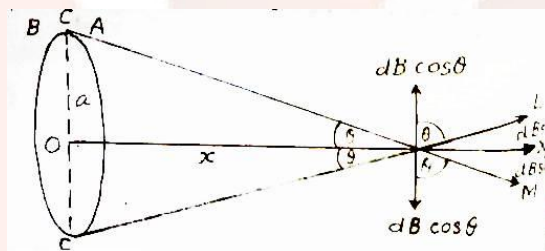
$$n_1 \sin i = n_2 \sin r$$

14. Explain Biot-Savart law. With its help derive an expression for the magnetic field at any point on the axis of a current carrying circular loop.

Answer: Consider a circular coil of radius 'a',

Centre O and current I is flowing from this loop,

According to Biot-Savart law,



The magnetic field at point P due to current element AB

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{Idl \sin 90^\circ}{(a^2 + x^2)^{3/2}}; \quad dB = \frac{\mu_0}{4\pi} \cdot \frac{Idl}{(a^2 + x^2)^{3/2}}$$

The effective component of magnetic field due to current element $dB \sin \theta$

The magnetic field at point P due to whole circular loop

$$B = \oint dB \sin \theta = \frac{\mu_0}{4\pi} \cdot \frac{I \sin \theta}{(a^2 + x^2)^{3/2}} \oint dl$$

Now, $\oint dl = 2\pi a$ circumference of loop

$$B = \frac{\mu_0}{4\pi} \cdot \frac{Ia \cdot 2\pi a}{(a^2 + x^2)(a^2 + x^2)^{1/2}}$$

$$B = \frac{\mu_0}{4\pi} \cdot \frac{Ia \cdot 2\pi a}{(a^2 + x^2)^{3/2}}$$

If the circular loop consists of n turns then

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi I a^2}{(a^2 + x^2)^{3/2}}$$

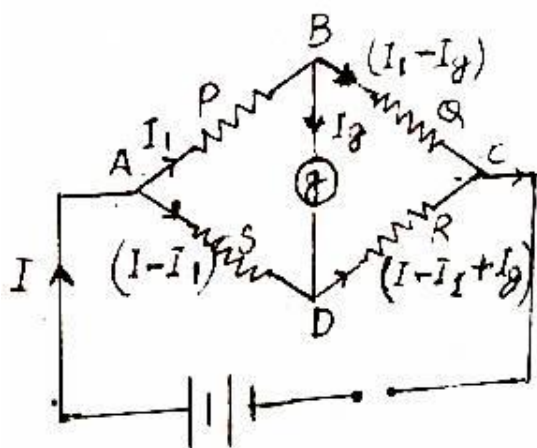
OR

Define Kirchhoff's laws of electrical circuit. Derive for expression for a balanced Wheatstone's bridge using Kirchhoff's law.

Answer: Kirchhoff's law: First law: In the network of conductor the algebraic sum of current meeting at a point is always zero. It is also called junction rule

Second law: It states that in any closed part of an electrical circuit, the algebraic sum of the emf is equal to the algebraic sum of the products of the resistance and current flowing through them. It is also called loop rule. This law is based upon the law of conservation of energy.

Wheatstone bridge: It is one of the accurate arrangements for measuring an unknown resistance. It is based on the principle of Kirchhoff's law.



Four resistance P, Q, R and S are connected to form a quadrilateral of galvanometer and a tapping key k_1 (called galvanometer) are connected between points B and D as shown in the figure.

Applying Kirchhoff's second law, for closed path ABDA in the circuit

$$I_1 P + I_g \cdot g - (I - I_1) S = 0$$

Where g is the resistance of galvanometer

In the balanced state $I_g = 0$

$$I_1 P + 0 \cdot g - (I - I_1) S = 0$$

$$I_1 P = (I - I_1) S$$

$$\frac{P}{S} = \frac{I - I_1}{I_1} \dots\dots\dots (i)$$

Applying Kirchhoff's law in closed part BDCB

$$I_g \cdot g + (I - I_1 + I_g) R - (I_1 - I_g) Q = 0$$

$$0 + (I - I_1) R - I_1 Q = 0$$

$$(I - I_1) R = I_1 Q$$

$$\frac{Q}{R} = \frac{I - I_1}{I_1}$$

From equation (i) and (ii) $\frac{P}{S} = \frac{Q}{R}; \quad S = \frac{P}{Q} \cdot R$

Hence, unknown resistance can be calculated.

15. Give postulates of Bohr's theory, Explain hydrogen spectrum on the basis of Bohr's theory.

Answer:

Various postulates of Bohr's atomic model are:

1. In an atom, the electrons revolve around the nucleus in certain definite circular paths called orbits, or shells.
2. Each shell or orbit corresponds to a definite energy. Therefore, these circular orbits are also known as energy levels or energy shells.
3. The orbits or energy levels are characterized by an integer n , where, n can have value 1, 2, 3, 4, The integer n ($=1, 2, 3, \dots$) is called the quantum number of respective orbit. The orbits are numbered as 1, 2, 3, 4, etc., starting from the nucleus side. Thus, the orbit for which $n=1$ is the lowest energy level. The orbit corresponding to $n=1, 2, 3, 4, \dots$ etc., are also designated as K, L, M, N etc. shells. When the electron is in the lowest energy level, it is

said to be in the ground state. Since, electrons can be these energy levels, electrons in an atom can have only certain permissible energies.

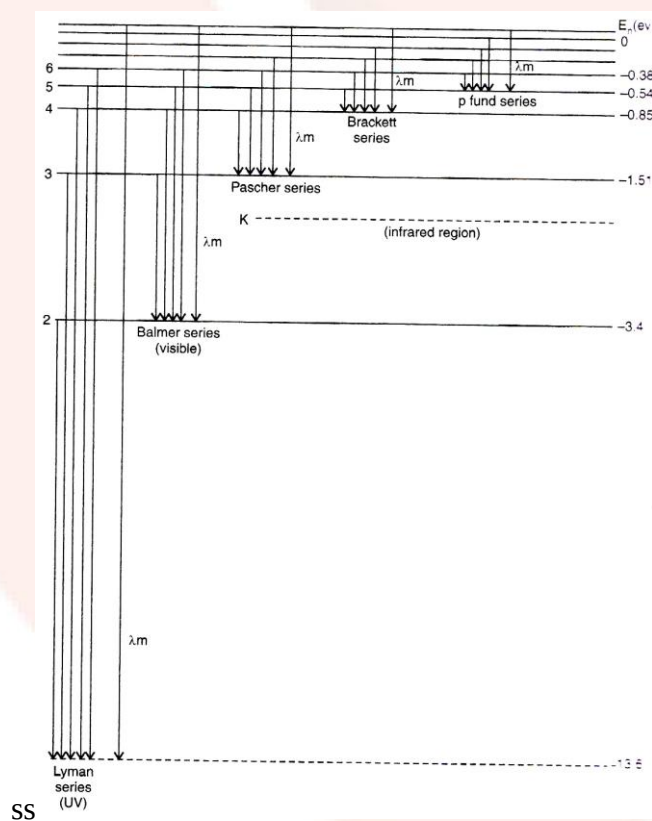
- The electrons present in an atom can move from a lower (E_{lower}) to a level of higher energy (E_{higher}) to a lower energy (E_{lower}) by losing the appropriate energy.

Hydrogen spectrum: Hydrogen emission spectrum consists of following series.

(i) **Lyman series:** This series is produced when electron jumps from high orbits to the first stationary orbit ($n_f = 1$)

$$\frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{n_i^2} \right] \text{ where } n_i = 2, 3, 4, 5$$

This series lies in ultraviolet region.



- (i) **Balmer series:** The series is produced when an electron jumps from higher orbit to second stationary orbit

$$\frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n_i^2} \right] \text{ where } n_i = 3, 4, 5, 6$$

- (ii) **Paschan series:** This series is produced when an electron jumps from higher orbit to third stationary orbit ($n_f = 1$)

$$\bar{\nu} = \frac{1}{\lambda} R \left[\frac{1}{3^2} - \frac{1}{n_i^2} \right] \quad n_i = 4, 5, 6, 7$$

- (iii) **Brackett series :** This series is produced when an electron jumps from higher orbit to the fourth stationary orbit ($n_f = 1$)

$$\bar{\nu} = \frac{1}{\lambda} R \left[\frac{1}{4^2} - \frac{1}{n_i^2} \right] \quad \text{Where } n_i = 5, 6, 7, 8$$

- (iv) **P fund series:** The series is produced when an electron jumps from higher orbit to the fifth stationary orbit ($n_f = 1$)

$$\bar{\nu} = \frac{1}{\lambda} R \left[\frac{1}{5^2} - \frac{1}{n_i^2} \right] \quad \text{Where } n_i = 6, 7, 8$$

This series are found in infrared region.

OR, Write down Einstein's photoelectric equation and explain the photoelectric effect on its basis.

Answer: Let m is the mass of an electron and v , its velocity then

Energy of the incident photon = work function + kinetic energy of the electron

$$h\nu = w + \frac{1}{2}mv^2 \dots\dots\dots(1)$$

If the electron does not lose energy by internal collisions, as it escape from the metal, the entire energy ($h\nu = w$) will be exhibited as the kinetic energy of electron.

Thus, ($h\nu = w$) represents the maximum kinetic energy of the ejected photo electron. If $V_{\max}$ is the maximum velocity with which the photoelectron can be ejected, then

$$h\nu = w + \frac{1}{2}mv^2(\max) \dots\dots\dots(2)$$

This equation is known as **Einstein's** photoelectric equation.

When the frequency (ν) of the incident radiation is equal to the threshold frequency (ν_0) of the metal surface, kinetic energy of the electron is zero.

Then equation (2) becomes,

$$h\nu_0 = w \dots\dots\dots(3)$$

Substituting the value of **W** in equation (2) we get.

$$h\nu - h\nu_0 = \frac{1}{2}mv^2 (\text{max})$$

$$h(\nu - \nu_0) = \frac{1}{2}mv^2 (\text{max})$$

This is another form of Einstein's photoelectric equation.

Photoelectric effect: when the light of energy hF falls on a metal surface. Then electron will absorb this energy and this will be used in following two ways:

- (i) A part of energy is used to overcome surface barrier and electron will come out from it. This part of energy is called work function.

$$\phi_0 = hf_0$$

- (ii) Remaining part of energy is used in giving a velocity v to the emitted photoelectron.

$$\frac{1}{2}mv^2 (\text{max})$$

Law of conservation of energy

S